

Scalability & Future Plan

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Current Limitations

- Single-model, single-prediction-type system (approve/reject only).
- No persistent database; each request is stateless.
- Not integrated with any real banking data source.
- Team capacity is currently limited to 2 active contributors.

Scalability Plan

- Move from a local Flask dev server to a production WSGI server (e.g. Gunicorn) behind Nginx.
- Containerize the application with Docker for consistent deployment across environments.
- Introduce a lightweight database to log predictions for monitoring and future model retraining.
- Add caching for repeated identical requests to reduce redundant computation.

Future Functional Enhancements

- Loan amount recommendation in addition to approve/reject classification.
- Model explainability (e.g. SHAP/LIME) so users understand which factors drove their result.
- Periodic model retraining pipeline as new (anonymized) data becomes available.
- Role-based views: a simplified view for applicants, a detailed view for loan officers.

Team Scaling Plan

Once Durgam Jyothrimai and Atte Archana are confirmed and onboarded, remaining unassigned stories (Multivariate Analysis, Pre-processing, and parts of Model Building) can be distributed to parallelize development. Boys Sai Yaswanth's tasks will be reassigned once device access is resolved.